

Fake News Detection in Indic Context: using Retrieval-Augmented Generation and Contrastive Learning Frameworks

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Abstract

Fake news detection in the Indic context presents distinct challenges such as low-resource settings and the scarcity of structured misinformation data. This work introduces a comprehensive framework addressing these challenges through two complementary architectures. First, a Retrieval-Augmented Generation (RAG) pipeline provides evidence-grounded predictions by leveraging a curated Indic-context dataset of fact-checked articles. Second, a novel Contrastive Learning for Fake News Detection (CLFD) model employs a dual encoder design with a cross attention bridge to jointly model semantic fidelity and stylistic distortion, capturing misinformation patterns beyond what retrieval alone can surface. A curriculum learning strategy with progressively harder augmentations, including LLM generated adversarial samples, further strengthens generalization. Together, the two architectures address complementary failure modes, offering a robust and interpretable solution for misinformation detection in low-resource Indic settings.

1 Introduction

The rapid proliferation of misinformation on digital platforms has emerged as a critical challenge, particularly in culturally diverse regions such as India (Sharma et al., 2019; Mustafaraj and Metaxas, 2017). The Indic context presents unique difficulties for automated fake news detection due to low-resource settings, regional linguistic variations, and the scarcity of high-quality annotated datasets (Nigam et al., 2021; Borgohain et al., 2023; Al-Zaman, 2022). Traditional approaches, largely developed for English, often fail to generalize effectively to the Indic context, resulting in reduced reliability and robustness in real-world deployments (Nigam et al., 2021).

Recent advances in large language models and retrieval-based systems have shown promise in

addressing misinformation by incorporating external knowledge sources (Sharma et al., 2019). Retrieval-Augmented Generation (RAG) frameworks enhance interpretability by grounding predictions in verifiable evidence (Sharma et al., 2019; Zhang et al., 2022). However, such approaches remain heavily dependent on the completeness and quality of the retrieval corpus, making them vulnerable in scenarios where relevant evidence is sparse, noisy, or unavailable—conditions that are common in low-resource Indic ecosystems (Nigam et al., 2021; Borgohain et al., 2023).

In parallel, representation learning techniques—particularly contrastive learning—have demonstrated strong potential in capturing nuanced semantic and stylistic patterns within text (Nigam et al., 2021; Khan et al., 2023). These methods enable models to distinguish subtle differences between factual reporting and misleading narratives, even in the absence of explicit external knowledge (Khan et al., 2023). Despite their effectiveness, the application of contrastive learning for fake news detection in Indic settings remains underexplored, particularly in conjunction with modeling stylistic distortion alongside semantic correctness (Nigam et al., 2021).

To address these challenges, we propose two complementary architectures. The first is a Retrieval-Augmented Generation (RAG)-based pipeline that leverages the structured corpus to produce evidence-grounded predictions, offering interpretability and explicit factual grounding. The second is a novel Contrastive Learning for Fake News Detection (CLFD) model, which employs a dual-encoder design coupled with a cross-attention bridge to jointly model semantic fidelity and stylistic distortion, capturing intrinsic misinformation patterns that may not always be surfaced through retrieval alone. Additionally, a curriculum learning strategy with progressively harder augmentations including LLM-generated adversarial samples is

introduced to improve generalization. Together, the two architectures address complementary failure modes: RAG excels when relevant evidence is retrievable, while CLFD provides robust signal when retrieval is incomplete or noisy.

Through extensive experiments, we demonstrate that both architectures contribute meaningfully to the task of Indic fake news detection, with their complementary strengths offering a more complete solution than either approach alone.

2 Related Work

The rapid proliferation of digital misinformation has prompted extensive computational, behavioral, and networks-based research. This section contextualizes our proposed architecture within existing methodologies, focusing on computational detection, natural language processing (NLP) applications, and the distinct socio-cultural challenges of the Indian information ecosystem.

2.1 Architectural Baselines

Automated veracity assessment historically relies on textual content or architectural security frameworks. Content-driven approaches have utilized structural statement conflict analysis to detect semantic anomalies between opposing text blocks (Zhang et al., 2022). Broad computational taxonomies categorize these methodologies into knowledge, style, propagation, and source-based detection pipelines (Sharma et al., 2019), while macro-level analyses emphasize building preventative platform architectures alongside standard reactive classification tools (Mustafaraj and Metaxas, 2017). Modern frameworks complement static text features by utilizing graph-based propagation models to identify influential network debunkers (Xu et al., 2022) or by capturing the co-evolutionary, temporal dynamics between true and false claims on social media platforms (Wang et al., 2022).

2.2 Deep Learning Integration

Transformer-based architectures and deep learning models have significantly advanced automated classification, moving beyond binary labels toward behavioral forensics that distinguish between deliberate disinformation agents and natural refutation spreaders (Khan et al., 2023). Concurrently, research has focused on integrating human-in-the-loop and hybrid oversight signals into neural training pipelines. The crowdsourcing paradigm categorizes human participants into distinct annotator,

evaluator, and creator subsets to serve as structured supervision data (He et al., 2025). Expanding on this crowd-driven framework, joint modeling architectures synthesize signals from both representative and synthetic human crowds to generate stable proxies for ground-truth labeling (t'Serstevens et al., 2024).

2.3 The Indian Information Landscape

Deploying NLP models in the Indian subcontinent introduces distinct domain-specific challenges, including extensive linguistic heterogeneity and complex regional context (Nigam et al., 2021). Thematic and qualitative reviews of localized misinformation during high-density public crises highlight recurring narratives centered on health claims, political manipulation, and religious polarization (Borgohain et al., 2023; Al-Zaman, 2022). Furthermore, computational analysis reveals that Indic disinformation is frequently entangled with deep ideological and communal frameworks (Das and Ahmed, 2022), making surface-level lexical models insufficient.

To counteract these dynamics, field experiments emphasize evaluating localized media literacy and educative interventions (Badrinathan, 2021). These interventions are conceptually structured around information behavior frameworks that establish definitions distinguishing between misinformation, disinformation, and malinformation across different creation, dissemination, and mitigation stages (Fernandez and Alani, 2018; Agarwal and Alsaedi, 2020, 2021).

2.4 Crisis Communication and Information Behavior

From a crisis communication perspective, systemic operational deficits often hamper real-time monitoring and mitigation workflows within high-density environments (Mehta et al., 2021). To formalize these vulnerabilities, information behavior frameworks model the progression of deceptive content across distinct creation, dissemination, and consumption phases (Agarwal and Alsaedi, 2020, 2021). Platform-level structural design flaws can further amplify these streams, exposing users to sustained persuasion effects that distort downstream public behavior and alignment (Mustafaraj and Metaxas, 2017; Toledano et al., 2024).

3 Dataset

We evaluate our proposed models on the **Bharat-Gather**¹ benchmark dataset (?), a specialized corpus engineered to capture domain-specific misinformation within the context of Indian mass gatherings (e.g., religious festivals, political rallies, and cultural events). Rather than relying on domain-general or Western-centric datasets, all instances are grounded in empirical data compiled from prominent Indian fact-checking platforms—e.g., *AltNews*, *BoomLive*—and enriched via multimedia YouTube transcript extractions and controlled synthetic text generation.

Every record conforms to a standardized eight-attribute schema comprising: `id`, `headline`, `src`, `label`, `training_text`, `full_article`, `url`, and `content_type`, ensuring structural consistency across all acquisition tiers.

4 Experiments

4.1 Data Splits and Input Formatting

For the Contrastive Learning for Fake News Detection (CLFD) model, 5% of the training split was withheld through stratified sampling to create an internal hold-out set. Furthermore, the test split was partitioned such that 40% was used for validation, while the remaining 60% was reserved for final evaluation. This setup intentionally exposes the model to validation samples drawn from unseen test-domain characteristics, thereby encouraging stronger generalization rather than overfitting to training-domain patterns. Each input sample was constructed by concatenating the article headline and training text. This formatting enables the model to capture both summary-level and contextual linguistic signals from the news articles.

For the RAG-based pipeline, the *BharatGather* dataset is transformed into a unified retrieval corpus by structurally conditioning heterogeneous multi-source news articles. The dataset is partitioned into training/indexing, validation, and testing splits for retrieval and inference-time evaluation.

Each article is converted into a structured semantic representation containing a **Claim**, **Verified Fact**, **Supporting Evidence**, **Verdict Label**, and **Metadata**. The verdict labels are drawn from the set {True, False, Misleading, Partly True, Unverified}, while metadata preserves source URLs and publisher information for traceability.

This structured format forms the retrieval corpus used for semantic matching during inference.

This structured representation forms the retrieval corpus against which incoming claims are semantically matched during inference.

4.2 Proposed Architectures

Two independent architectures are proposed for the task of fake news detection in the Indic context: a Retrieval-Augmented Generation (RAG) based pipeline and a Contrastive Learning for Fake News Detection (CLFD) model. The following subsections describe each in detail.

4.2.1 Architecture 1: RAG-Based Fake News Detection Pipeline

The proposed architecture employs a bifurcated Retrieval-Augmented Generation (RAG) framework designed to mitigate hallucination risks through evidence grounding within a curated knowledge base. The pipeline comprises two primary components: a deterministic Data Extraction Module and a probabilistic Grounded Inference Engine. As illustrated in Figure 1.

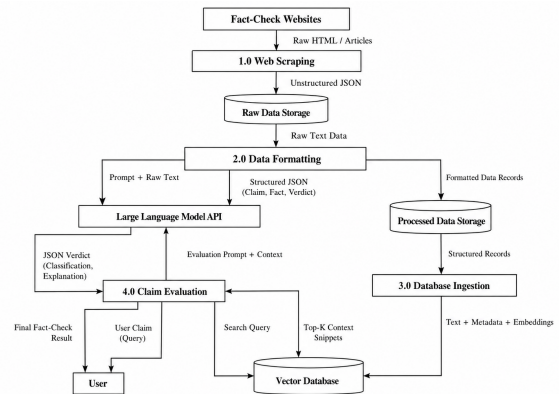


Figure 1: Proposed RAG-based architecture. The workflow proceeds from unstructured multi-source data ingestion to structured extraction, culminating in evidence-grounded claim verification.

Knowledge Base Construction To ensure data integrity and semantic coherence, raw news corpora ingested from regional Indic fact-checking platforms undergo structured synthesis via a zero-shot information extraction protocol. We employ the Qwen-3-32B large language model (LLM) constrained through a task-specific JSON schema (see Listing 1) to systematically extract high-density evidentiary triplets of the form (Claim, Fact, Evidence). This structured representation enables

¹BharatGather Dataset (13cube-pune/BharatGather)

the construction of a queryable knowledge graph wherein each fact-checked article is decomposed into its constituent semantic components.

To facilitate efficient indexing and maximize retrieval recall, the structured articles alongside their raw text undergo a rigorously empirical document chunking process. Specifically, long article text is algorithmically split into segments of 220 words with a 40-word overlap, constrained to a maximum of 12 chunks per article to balance computational overhead with context preservation. Both the dense fact extraction summaries (comprising Claim, Fact, Evidence, and Headline) and the raw article chunks are uniquely hashed utilizing a SHA-256 signature (combining the Source URL and the Claim) to prevent duplicate entries and maintain strict database hygiene. The resulting structured components are embedded and indexed into a persistent ChromaDB vector store utilizing the default all-MiniLM-L6-v2 embedding function framework. This architectural choice structurally binds multi-modal regional context to high-fidelity, unified semantic vectors, mitigating information loss in sparse domains.

Evidence-Grounded Inference During the inference stage, an incoming user claim serves as the prompt query vector. The ChromaDB instance is interrogated using L2 semantic similarity retrieval, extracting the top $k = 3$ most contextually relevant documents aligned with the query, alongside their metadata (such as source URLs and authoritative verdicts).

While the BharatGather dataset provides highly structured factual grounding and deterministic binary labels, it lacks synthesized, user-ready natural language explanations. To bridge this gap for real-world application, the retrieved evidentiary documents are concatenated and passed to a specialized “Truth in Crowd” generative agent. This agent, powered by the instruction-tuned Qwen 3 (32B) model operating at a temperature of 0.0, acts as a synthesizer that formats raw retrieved facts into an accessible, logical response.

To ensure the final output reads as an authoritative fact-check rather than a raw search result, the agent is governed by “negative constraints.” Specifically, the prompt explicitly forbids the use of internal system jargon or meta-discursive phrases (e.g., “according to the context,” “based on retrieved documents,” or “from our database”). This is further enforced through regex-based post-generation fil-

tering. As a result, the inference engine reliably yields a structured JSON payload (see Listing 2) mapping to four core components: (i) a categorical classification drawn from a discrete set {True, False, Misleading, Unverified}, (ii) a declarative formulation of the verified fact, (iii) an accessible natural-language explanation anchored strictly in the retrieved context, and (iv) an algorithmic confidence probability (0–100) reflecting the quantitative evidentiary strength. In scenarios where the language model fails to emit a confidence probabilistic score, a rule-based fallback matrix parameterized by the categorical classification governs the assignment. Finally, source URLs derived directly from the vector store metadata are iteratively bubbled up to the end-user, structurally permitting transparent source validation.

```
{
  "role": "Fact-Check Data Engineer",
  "task": "Transform raw news into high-density JSON.",
  "schema": {
    "claim": "Core assertion with subject and action.",
    "fact": "Standalone descriptive sentence of truth.",
    "evidence": "3-5 bullet points (dates, names, methods).",
    "verdict": "Strict: [True, False, Misleading, Partly True, Unverified]"
  },
  "output_constraint": "Return ONLY valid JSON."
}
```

Listing 1: System prompt for structured data extraction and semantic densification.

```
{
  "role": "Expert Fact-Checker",
  "instructions": [
    "Examine provided context for verification.",
    "Avoid meta-references like 'according to documents'.",
    "Provide plain, natural language explanation.",
    "Assign confidence score (0-100).",
  ],
  "output_keys": [
    "classification", "fact", "evidence",
    "explanation", "confidence", "sources"
  ]
}
```

Listing 2: Inference schema for the expert-level fact-checking agent.

4.2.2 Architecture 2: Contrastive Learning for Fake News Detection (CLFD)

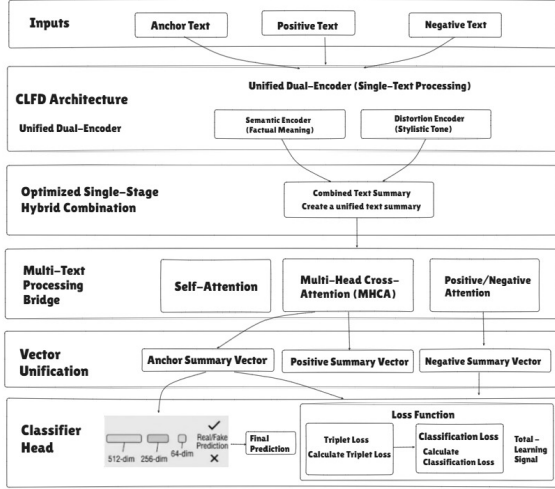


Figure 2: Proposed block diagram.

The second architecture is the Contrastive Learning for Fake News Detection (CLFD) model. This model assumes that fake news can be identified by looking at two things simultaneously: the actual facts being claimed, and the emotional or stylistic tone used to present them. To capture both, the model uses two parallel neural networks that work together, followed by an attention bridge and a final classifier.

Dual Encoder Processing for Triplets. During training, the system looks at groups of three texts (triplets): an original text (anchor), a factually equivalent paraphrase (positive), and a text where the tone is maliciously manipulated (negative). Crucially, *every* text in this triplet is processed independently through both encoders in the architecture.

The first encoder (RoBERTa_{base}) acts as the **semantic anchor**. Its job is to focus purely on the factual meaning of the text. The second encoder (BERT_{base-uncased}) acts as the **distortion detector**. Its job is to spot stylistic manipulation and emotional amplification, such as panic-inducing language.

To ensure the model captures both the high-level summary of the article and the specific vocabulary used, each encoder combines its outputs using a hybrid approach. It takes the overall sequence embedding (the [CLS] token) and averages it with the individual meanings of all other tokens (mean pooling). By averaging these together, the sequence length is removed, and the text is compressed down

into a single 256-dimensional summary vector per encoder.

Cross-Attention Contextual Bridge. Once the text is compressed into 256-dimensional vectors, the model needs to understand how the tone relates to the facts. For example, dramatic language might be completely justified if the facts describe a real disaster, but suspicious if the underlying facts are mundane.

To figure this out, we use a Multi-Head Cross-Attention (MHCA) bridge. Think of this process as the tone detector “asking questions” to the fact checker. The distortion vector acts as the *Query*, while the semantic vector acts as the *Key* and *Value*. By cross-referencing the tone against the facts, the network produces an “informed distortion” vector.

Finally, we glue this informed distortion vector back together with the original semantic vector, resulting in a single comprehensive 512-dimensional vector. This entire process generates the 512-dimensional vector independently for the anchor, positive, and negative inputs to compute the model’s contrastive loss.

Classifier Head Assembly. This final 512-dimensional vector is explicitly passed through a standard classification head (a series of shrinking dense layers reducing from 512 to 256, then 64, and finally down to 2 neurons). This outputs the final prediction, determining whether the text is classified as “real” or “fake.”

4.3 Training Strategy

4.3.1 Curriculum Learning

We train the model using “curriculum learning,” which means we start by giving it easy examples and gradually make them harder over eight training epochs. This happens in three stages:

Stage 1: Easy (Epochs 1-2). We create simple training examples by simply swapping words with their synonyms. To create fake news examples, we automatically insert dramatic words (like “panic” or “shocking”) into the text.

Stage 2: Medium (Epochs 3-4). We make the factual examples harder by translating the text into Hindi and then back into English. This creates natural-sounding paraphrases. Using Hindi as a middle step works especially well for our dataset because it keeps Indian names and places intact better than translating through non Indic language.

Stage 3: Hard (Epochs 5-8). For the hardest fake

news examples, we use a Large Language Model (Qwen-3-32B) to rewrite real news. We instruct the LLM to keep all the facts, names, and numbers exactly the same, but to change the tone maliciously to sound like misinformation. This trains the model to spot subtle emotional manipulation even when the core facts look correct.

4.3.2 Freeze Schedule and Optimizer

When training begins, the newly added classification layers just use random weights. If we train everything at once, these random early guesses could ruin the pre-trained knowledge in our encoders. To prevent this, we train the model in phases by gradually “unfreezing” parts of it:

Phase 1 (Epochs 1-2): Both the semantic and distortion encoders are completely frozen. The model only trains its new classification head and the attention bridge.

Phase 2 (Epochs 3-4): We unfreeze the distortion encoder so it can actively start learning how to detect emotional manipulation.

Phase 3 (Epochs 5-8): We finally unfreeze the semantic encoder, but we use a severely reduced learning rate (25% of the normal rate). This allows it to make minor adjustments to its fact-checking abilities without forgetting the core language patterns it already knows.

Every time we move to a new phase, we completely reset the optimizer. This ensures the training process is smooth and doesn’t carry over any outdated training momentum from when the layers were still frozen.

4.3.3 Loss Function

The model learns by keeping track of two different types of errors at the same time:

1. Triplet Loss (Contrastive Learning): Instead of just evaluating one article independently, the model compares the anchor, the positive (real paraphrase), and the negative (dramatic fake news) all at once. The goal is to mathematically pull the anchor’s meaning and the positive’s meaning closer together, while pushing the negative’s meaning at least a set distance away (a margin of 0.5).

To make the model even sharper, we use “Online Hard Negative Mining.” This means during training, the model scans the current batch of texts to find the trickiest, most confusing fake news example, and focuses specifically on pushing that “hard” example away. This forces the model to learn subtle differences rather than just easy mistakes.

2. Classification Loss (Cross-Entropy): This is a standard check to see if the model ultimately labeled the article as “fake” or “real” correctly. We use “label smoothing,” which prevents the model from becoming overly confident in its predictions—this helps it perform better on entirely new data. We also automatically balance the learning weights to ensure the model doesn’t just blindly guess the most common label if the dataset is unevenly distributed.

The final learning signal is a simple blend of these two error scores. We give 60% importance to getting the final “fake or real” classification right, and 40% importance to the triplet loss that helps shape the model’s underlying separation of facts and emotions.

4.4 Experimental Setup

4.4.1 Hyperparameters

The key hyperparameters for CLFD are summarized in Table 1.

Table 1: CLFD Hyperparameter Configuration

Hyperparameter	Value
Semantic encoder	RoBERTa-base
Distortion encoder	BERT-base-uncased
Max sequence length	128 tokens
Projection dimension	256
Cross-attention heads	4
Batch size (effective)	32 (8 × 4 steps)
Learning rate	2×10^{-5}
Semantic enc. LR (Ph. 3)	5×10^{-6}
Epochs	8
Dropout	0.3
Contrastive temperature	0.3
Triplet margin	0.5
Loss weight α	0.4
Early stopping patience	3 epochs

5 Results

5.1 Evaluation Metrics

Both architectures are evaluated using the macro-averaged F1 score as the primary metric, which handles class imbalance by weighting classes equally. Supporting metrics include accuracy, ROC-AUC, Matthews Correlation Coefficient (MCC), and per-class precision/recall. Calibration analysis is also performed for the CLFD model to evaluate confidence reliability.

5.2 RAG-Based Pipeline Performance

The Retrieval-Augmented Generation (RAG) pipeline shows strong performance in a low-

resource, retrieval-dependent setting.

Table 2: RAG Pipeline Performance

Metric	Value
Accuracy	90.91%
Macro F1 Score	0.91
Precision (macro)	0.93
Recall (macro)	0.90

The classification report reveals perfect precision for the false class but lower recall (0.80), indicating some false instances are misclassified as true. The true class achieves perfect recall, showing robustness in detecting genuine news. This reflects a conservative bias: the model validates claims with strong retrieval evidence but struggles with weak or ambiguous contexts.

5.3 CLFD Model Performance

The Contrastive Learning for Fake News Detection (CLFD) model outperforms the RAG pipeline across all metrics.

Table 3: CLFD Model Performance

Metric	Value
Accuracy	97.82%
Macro F1 Score	0.9777
ROC-AUC	0.9945
MCC	0.9553

The high ROC-AUC demonstrates excellent class separability, while the MCC confirms robust performance under imbalance. The near-perfect macro F1 score reflects balanced precision and recall.

Table 4: Comparison of Architectures

Metric	RAG	CLFD
Accuracy	90.91%	97.82%
Macro F1	0.91	0.9777
Robustness to Style Variations	Moderate	High
Dependency on External Knowledge	High	Low
Generalization Capability	Moderate	Strong

RAG depends on retrieval quality, offering explainability via evidence but faltering without relevant facts. CLFD, by contrast, learns intrinsic misinformation patterns through contrastive learning, curriculum training, and cross-attention, enabling

superior generalization to unseen data and stylistic variations.

6 Conclusion

This work introduced a dual-architecture misinformation detection framework tailored to the socio-cultural and low-resource challenges of the Indic information ecosystem using the **BharatGather** dataset (?). Experimental results demonstrate that both the Retrieval-Augmented Generation (RAG) pipeline and the Contrastive Learning for Fake News Detection (CLFD) model contribute complementary strengths toward reliable misinformation detection. The RAG pipeline provides strong factual grounding and explainability through external evidence retrieval, achieving an accuracy of 90.91%. In parallel, the CLFD model captures intrinsic semantic consistency and stylistic distortions within content, achieving an accuracy of 97.82% and an ROC-AUC of 0.9945. Together, these architectures address different operational challenges: RAG performs effectively in evidence-rich scenarios, while CLFD offers robust generalization in noisy information environments.

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Appendix

This appendix provides the verbatim prompt strings. These prompts are designed for the Qwen-3-32B model to ensure consistent JSON formatting and evidentiary grounding.

Extraction Prompt (Preprocessing)

The following prompt is utilized by the *Data Extraction Module* to transform unstructured news text into a structured knowledge base.

```
You are a Fact-Check Data Engineer. Your
task is to transform raw news
articles into structured, high-
density JSON for a misinformation
detection RAG system.

### EXTRACTION RULES:
1. "claim": Extract the core assertion.
   Include the subject and the alleged
   action.
2. "fact": Write a standalone,
   descriptive sentence of the verified
   truth.
3. "evidence": Provide 3-5 specific
   bullet points. Include dates, names
   of authorities, or technical methods
   used for verification.
4. "verdict": Strictly use one of these:
   [True, False, Misleading, Partly
   True, Unverified].

### OUTPUT FORMAT:
Return ONLY a valid JSON object. Do not
include any introductory or
concluding text.

### YOUR TASK:
Analyze the following data:
{article_data}

Output JSON:
```

Listing 3: Verbatim Extraction Prompt

Inference Prompt (Expert Agent)

The following prompt governs the *Grounded Inference Engine*. It enforces negative constraints to

ensure the agent does not reference the underlying database structure in its user-facing explanation.

```
Act as an expert fact-checker for the "
Truth in Crowd" misinformation
detection system. Your task is to
evaluate the user's claim strictly
based on the provided context
retrieved from our database.

Context Documents:
{context_str}

User Claim to Verify:
"{user_claim}"

Instructions:
1. Examine the provided context for
   facts to verify or debunk the claim.
2. Determine if the claim is True, False
   , Misleading, or Unverified.
3. Provide a plain, natural explanation
   for the user.
4. Do NOT mention words like "context",
   "retrieved documents", or "database"
   in the explanation.
5. Assign a confidence score from 0 to
   100 that reflects how strongly the
   context supports the verdict.
6. Output your response entirely in JSON
   format with the following keys: ["
   classification", "fact", "evidence",
   "explanation", "confidence", "
   sources"].
```

Listing 4: Verbatim Inference Prompt